

descSPIM-Fullmoon: Python Control Instructions

These instructions outline the procedure for controlling the descSPIM-Fullmoon using Python to perform large sample imaging.

※This guide is intended for use with Windows 11.

※Computer specs require at least 64GB of RAM.

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1. Environment creation

1-1. Application Installation

Install the required environment from the following URLs.

1-1-1. Installation of Python

<https://www.python.org>

Download the installer and follow the instructions to install Python. Select "Add Python to PATH" during installation and add Python to the PATH environment variable.

Currently, this PC is using Python 3.13.9. Different versions and bit numbers may lead to errors.

1-1-2. Installation of Nikon DS50M-SDK

<https://nisdk.recollective.com>

Please apply for permission to use the "DS50M SDK" from the "Sign-up here". It may take several weeks for the application to be approved and for the SDK to become available.

Once approved, download and install the "DigitalSight50M SDK 64-bit".

The installation folder will be located at "C:\Program Files\Nikon".

1-1-3. Installation of XA software

https://www.thorlabs.co.jp/software_pages/ViewSoftwarePage.cfm?Code=Motion_Control&viewtab=0

Follow the installation guide to install XA Software for 64-Bit Windows from the above URL. The installation folder will be located at "C:\Program Files\Thorlabs XA".

Operation was tested with version 1.3.0.

1-1-4. Installation of ThorCAM software

https://www.thorlabs.co.jp/software_pages/ViewSoftwarePage.cfm?Code=ThorCam

Install ThorCam Software for 64-Bit Windows from the above URL. Follow the installation guide. Use version 3.7.0.6

1-2. Environment Setup

1-2-1. Creation of a new environment

Open Command Prompt and run the following commands one by one.

Check that Python is installed:

```
python --version
```

Create a project folder (if it does not already exist):

```
mkdir %USERPROFILE%\Projects\descSPIM
```

Move into the project folder:

```
cd %USERPROFILE%\Projects\descSPIM
```

Copy the files from GitHub to the project folder. Run the following commands one by one:

```
powershell -Command "Invoke-WebRequest -Uri  
https://github.com/dbsb-juntendo/descSPIM-Advanced/archive/refs/heads/main.zip -OutFile full.zip"
```

```
powershell -Command "Expand-Archive -Path full.zip  
-DestinationPath ."
```

```
xcopy descSPIM-Advanced-main\descSPIM-FullMoon descSPIM-Fullmoon  
/E /H /Y
```

```
rmdir /S /Q descSPIM-Advanced-main
```

```
del full.zip
```

```
cd %USERPROFILE%\Projects\descSPIM\descSPIM-Fullmoon
```

Create a virtual environment:

```
py -3.13 -m venv .myenv-descSPIM-Fullmoon
```

Activate the virtual environment:

```
.myenv-descSPIM-Fullmoon\Scripts\activate
```

Copy the XA.library:

```
copy "C:\Program Files\Thorlabs XA\SDK\Native (C,
C++)\Libraries\x64\tlmc_xa_native.dll"
"%USERPROFILE%\Projects\descSPIM\descSPIM-Fullmoon\"
```

1-2-2. Installation of required packages

Installation of xa_sdk_setup.zip into the project folder (%USERPROFILE%\Projects\descSPIM\descSPIM-Fullmoon). After activating the virtual environment, run the following pip commands one by one:

```
curl -L -o xa_sdk_setup.zip
https://github.com/Thorlabs/Motion_Control_Examples/raw/main/Python/XA/xa_sdk_setup.zip
```

```
pip install xa_sdk_setup.zip
```

Copy the thorCAM libraries into the project folder (%USERPROFILE%\Projects\descSPIM\descSPIM-Fullmoon).

After activating the virtual environment, run the following pip commands one by one:

```
powershell -Command "Expand-Archive -LiteralPath 'C:\Program
Files\Thorlabs\Scientific Imaging\Scientific Camera
Support\Scientific_Camera_Interfaces.zip' -DestinationPath ."
```

```
pip install
"%USERPROFILE%\Projects\descSPIM\descSPIM-Fullmoon\ThorlabsSDK\Sci
entific Camera Interfaces\SDK\Python
Toolkit\thorlabs_tsi_camera_python_sdk_package.zip"
```

```
mkdir
"%USERPROFILE%\Projects\descSPIM\descSPIM-Fullmoon\dlls\64_lib"
```

```
xcopy  
"%USERPROFILE%\Projects\descSPIM\descSPIM-Fullmoon\Scientific  
Camera Interfaces\SDK\Native Toolkit\dlls\Native_64_lib\*" "  
"%USERPROFILE%\Projects\descSPIM\descSPIM-Fullmoon\dlls\64_lib" /E  
/H /Y
```

Run the following commands one by one:

```
pip install PySide6
```

```
pip install numpy
```

```
pip install tifffile
```

```
pip install pyserial
```

```
pip install pywin32
```

```
pip install pythonnet
```

```
pip install jds6600
```

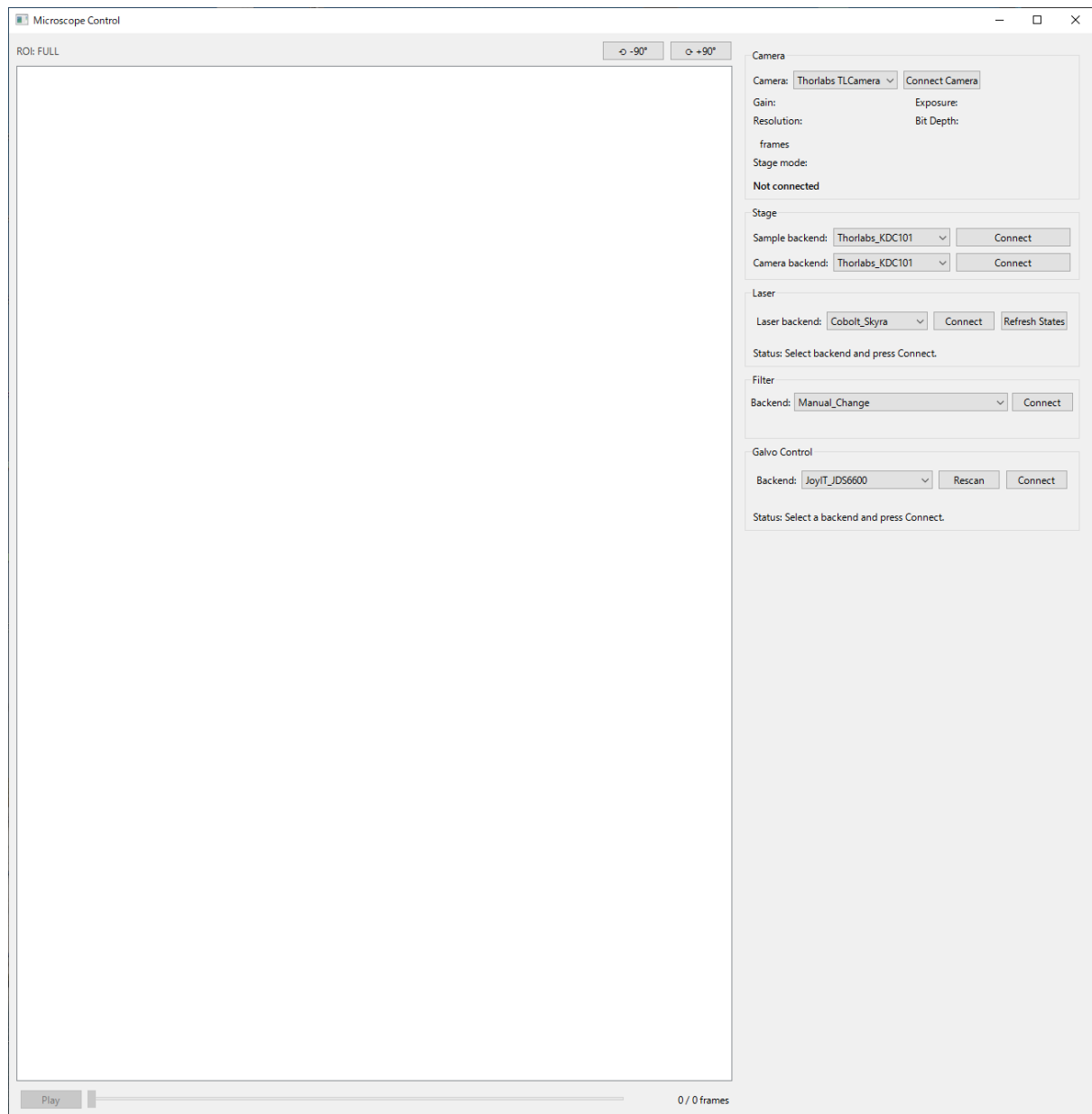
Run the following commands one by one:

```
cd  
%USERPROFILE%\Projects\descSPIM\descSPIM-Fullmoon\launcher
```

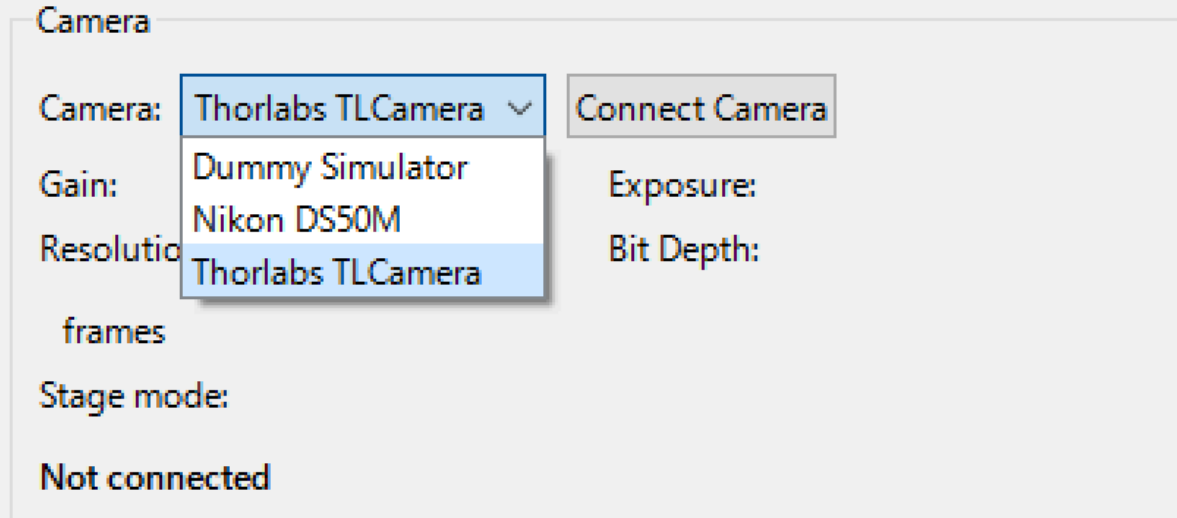
```
make_shortcut.cmd
```

2. Executing the Python code and using the GUI

After confirming that the devices are powered on and properly connected to the PC, double-click descSPIM icon on the desktop to launch the GUI.



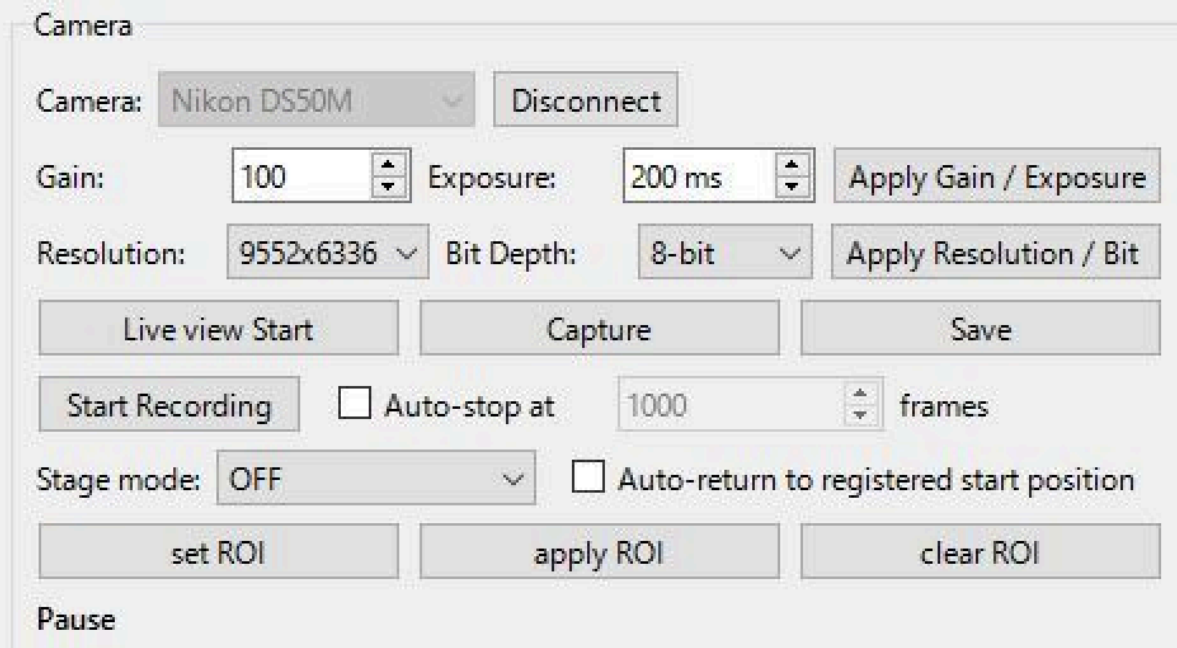
2-1. Camera control



The screenshot shows the 'Camera' control panel. At the top, the title 'Camera' is displayed. Below it, there is a 'Camera:' label followed by a dropdown menu that is currently open, showing three options: 'Thorlabs TLCamera' (selected), 'Dummy Simulator', and 'Nikon DS50M'. To the right of the dropdown is a 'Connect Camera' button. Below the dropdown, there are labels for 'Gain:', 'Resolution:', 'frames', 'Stage mode:', 'Exposure:', and 'Bit Depth:'. The 'Resolution:' label is partially obscured by the dropdown menu. At the bottom, the text 'Not connected' is displayed.

Select the backend for the connected camera from the combo box.

When you press the **Connect Camera** button, the camera connects and the camera control panel appears.



The screenshot shows the 'Camera' control panel after the camera has been connected. The title 'Camera' is at the top. Below it, the 'Camera:' label is followed by a dropdown menu showing 'Nikon DS50M'. To the right is a 'Disconnect' button. Below the dropdown, there are labels for 'Gain:', 'Exposure:', 'Resolution:', 'Bit Depth:', 'frames', 'Stage mode:', and 'Auto-stop at'. The 'Resolution:' label is partially obscured by the dropdown menu. Below these labels, there are buttons for 'Live view Start', 'Capture', and 'Save'. Below these buttons, there is a 'Start Recording' button, a checkbox for 'Auto-stop at', a spin box set to '1000', and the text 'frames'. Below this, there is a 'Stage mode:' label followed by a dropdown menu showing 'OFF', a checkbox for 'Auto-return to registered start position', and buttons for 'set ROI', 'apply ROI', and 'clear ROI'. At the bottom, the text 'Pause' is displayed.

To change the gain or exposure time, enter a value in the spin box and press Enter or the Apply Gain/Exposure button.

To change the resolution or bit depth, select an option from the pull-down menu and press the Apply Resolution / Bit button. (Some cameras may not support this function.)

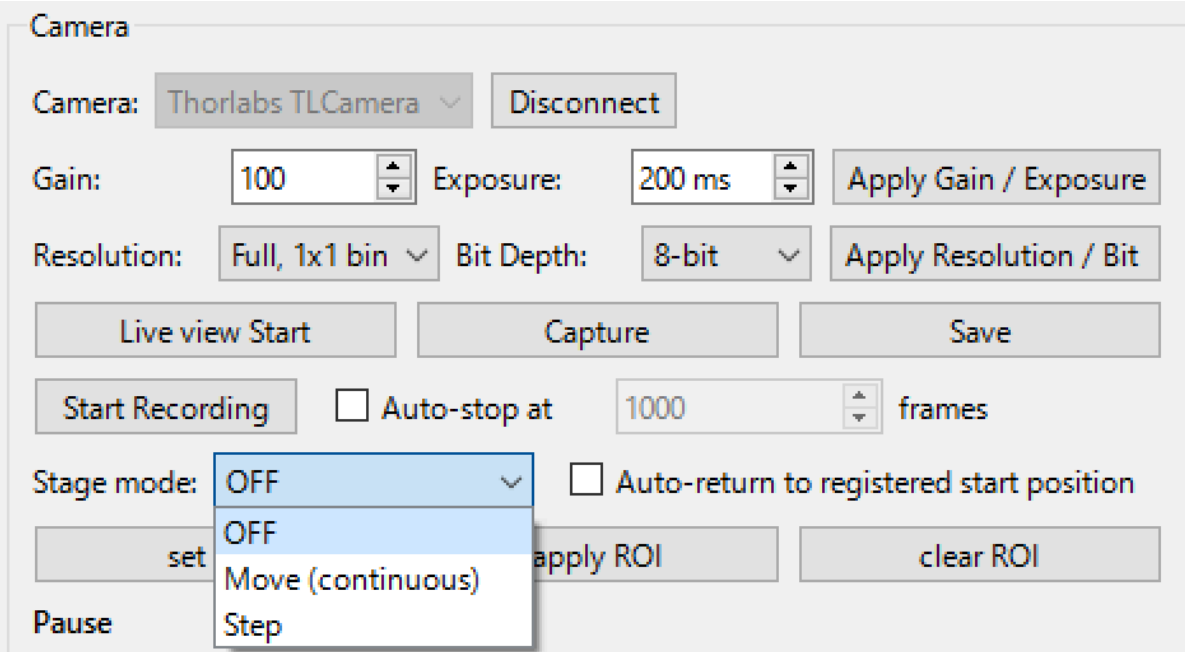
Press the Live View button to start live view.

Press the Capture button to take a snapshot.

Press Save to open a dialog for saving the snapshot.

To acquire data, press the Start Recording button.

If stages are connected, you can link the Start Recording button with stage movement.



The screenshot shows a software interface for camera control. At the top, there's a 'Camera' section with a dropdown menu set to 'Thorlabs TLCamera' and a 'Disconnect' button. Below this are input fields for 'Gain' (set to 100) and 'Exposure' (set to 200 ms), each with a small up/down arrow icon, followed by an 'Apply Gain / Exposure' button. The next row has 'Resolution' (set to 'Full, 1x1 bin') and 'Bit Depth' (set to '8-bit'), both with dropdown arrows, followed by an 'Apply Resolution / Bit' button. A row of three buttons follows: 'Live view Start', 'Capture', and 'Save'. Below these is a 'Start Recording' button, a checkbox for 'Auto-stop at' (unchecked), a value of '1000' with an up/down arrow, and the word 'frames'. The 'Stage mode:' dropdown is open, showing options: 'OFF' (selected), 'OFF', 'Move (continuous)', and 'Step'. To the right of this dropdown is another checkbox for 'Auto-return to registered start position' (unchecked). At the bottom left is a 'Pause' button. In the center bottom are 'set' and 'apply ROI' buttons. At the bottom right is a 'clear ROI' button.

Stage Mode:

OFF → No link with the stages

Move → Continuous movement of both stages during recording (movement follows the configured settings)

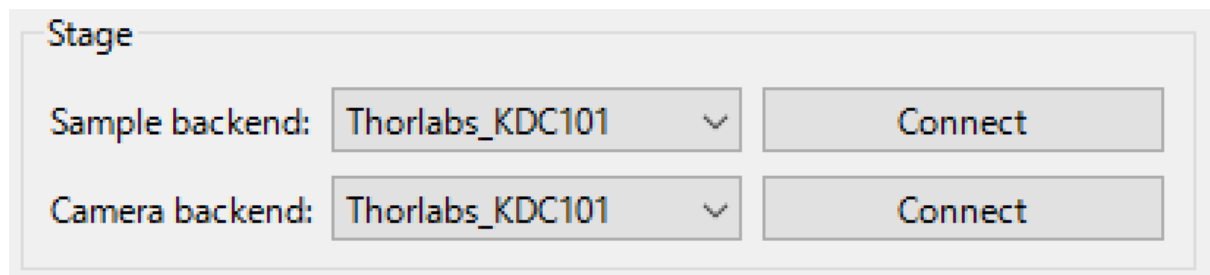
Step → Camera capture and step movement of both stages occur alternately (movement follows the configured settings)

When data recording stops, the stages also stop.

If you want the stages to automatically return to the registered start position, check Auto-return to registered start position.

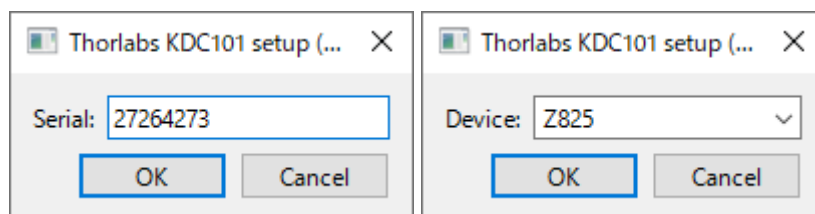
If a start position is registered, the stages automatically return to it. To enable automatic stopping of data acquisition, check Auto-stop at and specify the number of frames N to acquire.

2-2. Stage control

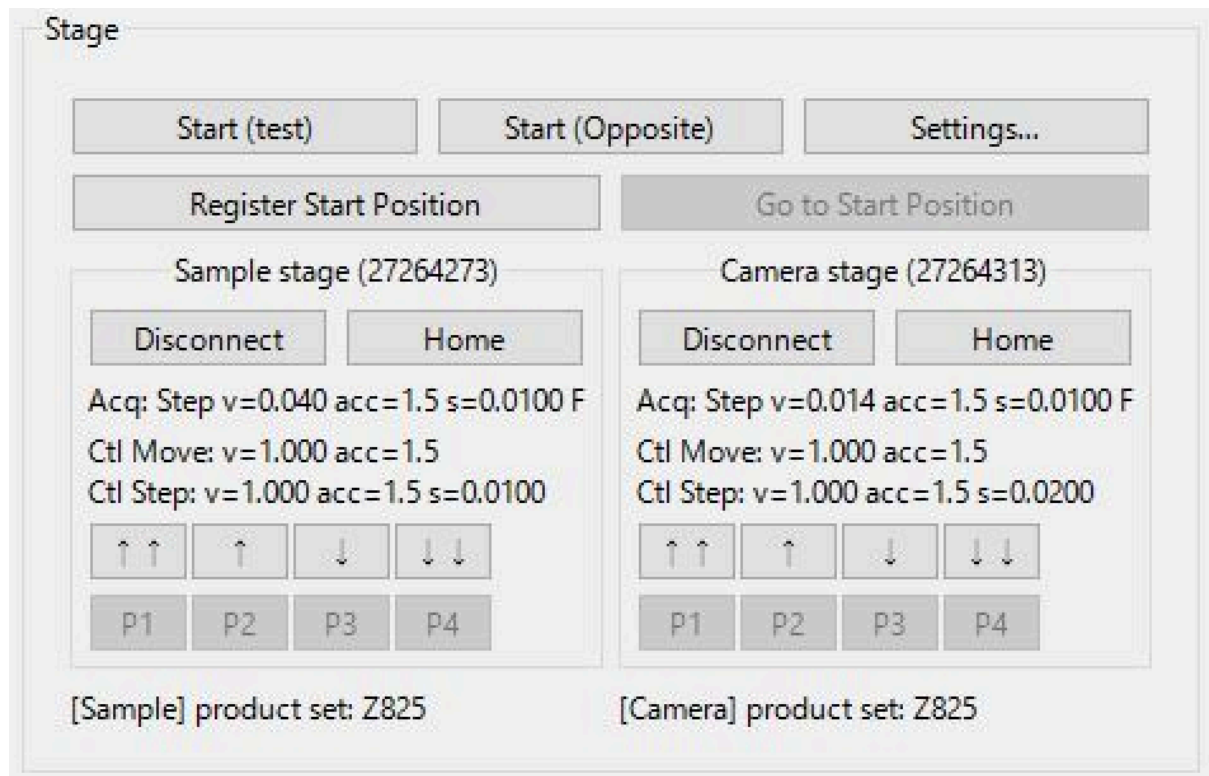


The screenshot shows a window titled "Stage". Inside, there are two rows. The first row is labeled "Sample backend:" and has a dropdown menu showing "Thorlabs_KDC101" and a "Connect" button. The second row is labeled "Camera backend:" and also has a dropdown menu showing "Thorlabs_KDC101" and a "Connect" button.

Select the backends for the connected sample stage and camera stage, and press the Connect button. A dialog will appear asking you to enter the information required for the connection. For example, with Thorlabs_KDC101, the serial number input dialog and the connected device input dialog will appear in sequence, so please enter the required information. The stages will connect and the stage control panel will appear. Stage connection takes about five seconds.



The image shows two sequential dialog boxes. The first dialog is titled "Thorlabs KDC101 setup (...)" and has a "Serial:" label followed by a text input field containing "27264273". It has "OK" and "Cancel" buttons. The second dialog is also titled "Thorlabs KDC101 setup (...)" and has a "Device:" label followed by a dropdown menu showing "Z825". It also has "OK" and "Cancel" buttons.



Start (test) button: Moves both actuators using the acquisition (Acq) settings. This allows you to test the motion that will be executed when recording starts.

Start (Opposite) button: Performs the same movement as Start (test), but in the opposite direction.

Setting... button: Opens the actuator settings dialog.

Register Start Position: Registers the current positions of the two stages. This is used to begin acquisition from the same position during multi-track (multi-color) imaging.

Go to Start Position button: Moves the stages to the registered start position.

Disconnect: Disconnects the controller.

Home: Moves the actuator to its home position (0).

Acq: Displays the current acquisition settings.

Mode (Move, Step) indicates the behavior when the Start (test) button is pressed, not the behavior used during recording.

Ctl Move: Displays the settings used for manual continuous movement (↑↑, ↓↓).

Ctl Step: Displays the settings used for manual step movement (↑, ↓).

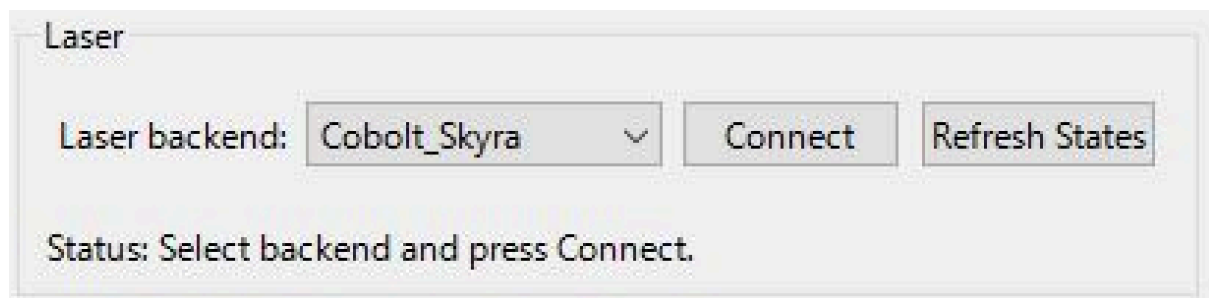
↑↑, ↓↓: Moves the actuators continuously in the Forward or Reverse direction.

↑, ↓: Moves the actuators in step mode in the Forward or Reverse direction.

Preset (1–4):

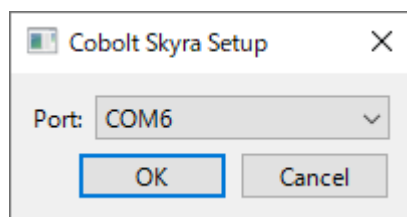
Switches between presets for manual operation settings.

2-3. Laser control



The screenshot shows a window titled "Laser". Inside, there is a label "Laser backend:" followed by a dropdown menu currently showing "Cobolt_Skyra". To the right of the dropdown are two buttons: "Connect" and "Refresh States". Below these elements, a status message reads: "Status: Select backend and press Connect."

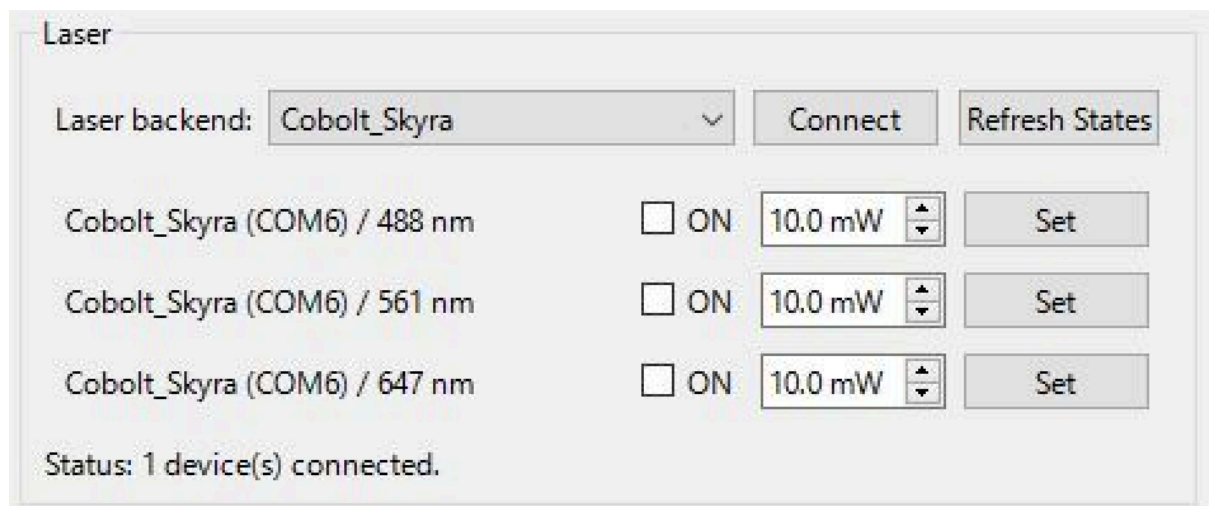
Select the backend for the connected laser and press the **Connect** button. A connection dialog will appear. If no COM port is shown, check the connection to the PC and press the **Refresh States** button in the dialog.



The screenshot shows a small dialog box titled "Cobolt Skyra Setup" with a close button (X) in the top right corner. It contains a label "Port:" followed by a dropdown menu showing "COM6". At the bottom, there are two buttons: "OK" and "Cancel".

Select the COM port to which the laser is connected, then press **Connect**. The laser will connect, and the laser control panel will appear.

For example, when a **Cobolt Skyra** is connected, the installed wavelengths are automatically detected and reflected in the control panel.



The screenshot shows the "Laser" window after successful connection. The "Laser backend:" dropdown still shows "Cobolt_Skyra". The "Connect" and "Refresh States" buttons are still present. Below them, three rows of laser configurations are listed:

Laser backend:	Wavelength	Power	Status	Action
Cobolt_Skyra (COM6)	488 nm	10.0 mW	☐ ON	Set
Cobolt_Skyra (COM6)	561 nm	10.0 mW	☐ ON	Set
Cobolt_Skyra (COM6)	647 nm	10.0 mW	☐ ON	Set

At the bottom, the status message reads: "Status: 1 device(s) connected."

Checking the ON/OFF checkbox next to each laser wavelength turns the laser on. To stop the laser, uncheck the box.

To change the laser power, enter a value in the spin box and press the **Set** button.

If you want to add another laser, you can select an additional backend file and press the **Connect** button to add a second or third laser.

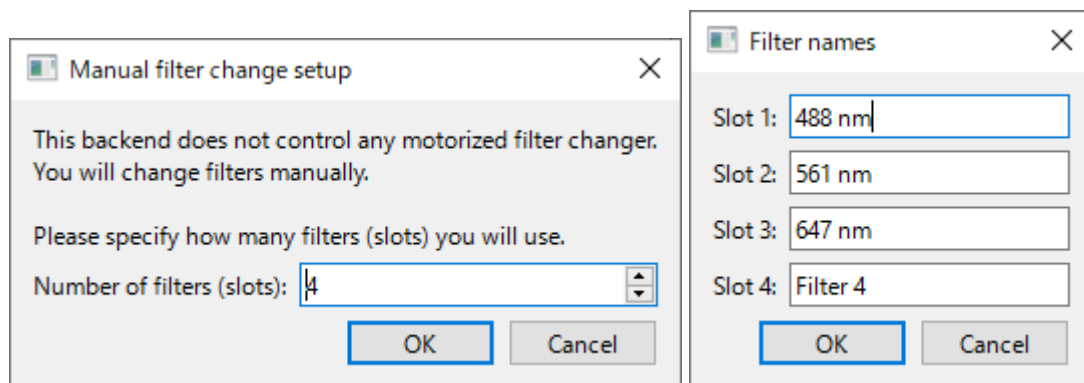
2-4. Filter control



Filter

Backend: Manual_Change ▼ Connect

Select the backend for the connected filter changer and press the **Connect** button. A connection dialog will appear. To replace the filter manually, select **Manual_Change**.



Manual filter change setup

This backend does not control any motorized filter changer. You will change filters manually.

Please specify how many filters (slots) you will use.

Number of filters (slots): 4

OK Cancel

Filter names

Slot 1: 488 nm

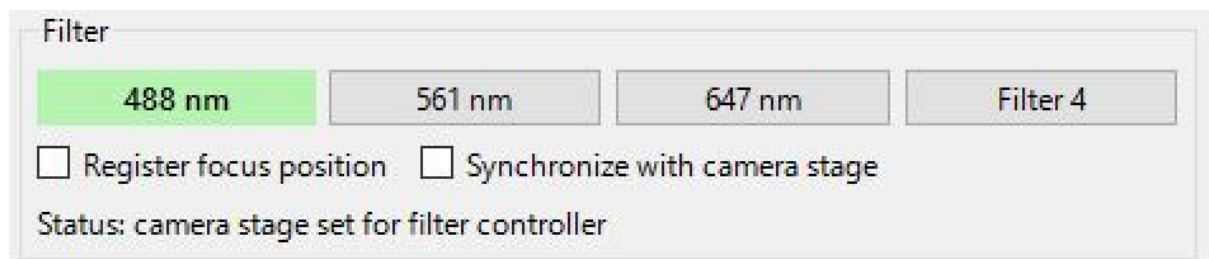
Slot 2: 561 nm

Slot 3: 647 nm

Slot 4: Filter 4

OK Cancel

Enter the required information (number of slots, slot names, etc.), then the filter changer will connect and the filter control panel will appear.



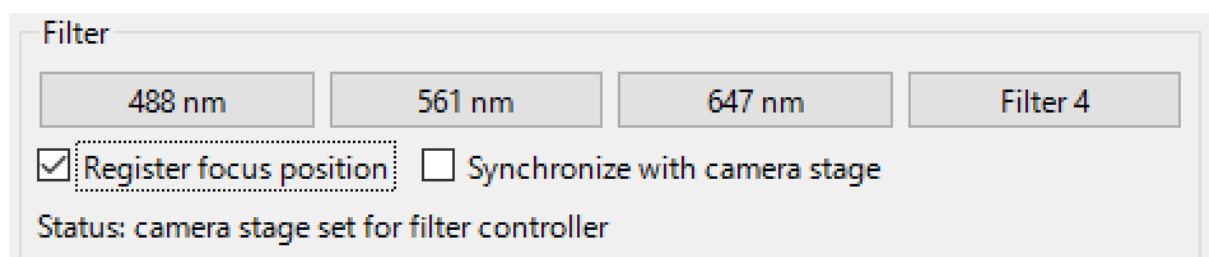
Filter

488 nm 561 nm 647 nm Filter 4

☐ Register focus position ☐ Synchronize with camera stage

Status: camera stage set for filter controller

Pressing a displayed slot name button moves the filter to that slot. To correct focus shifts caused by filter changes, you can record the corresponding focus positions.



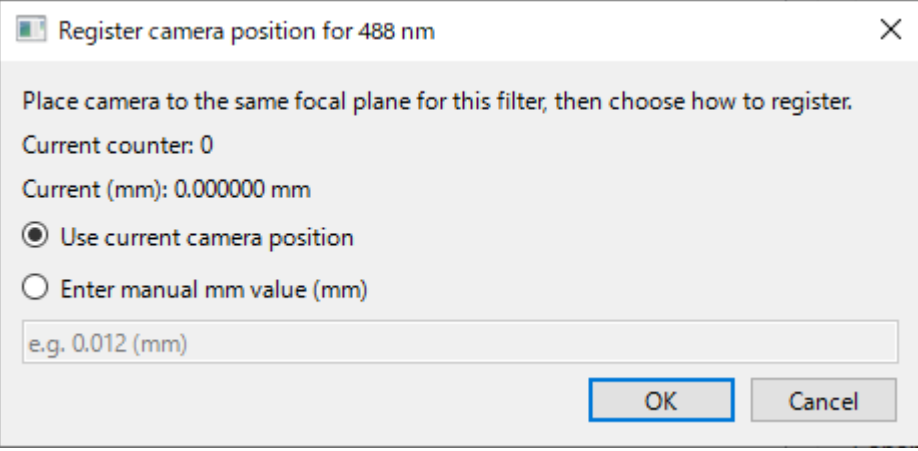
Filter

488 nm 561 nm 647 nm Filter 4

☒ Register focus position ☐ Synchronize with camera stage

Status: camera stage set for filter controller

When **Register focus position** is checked, pressing each slot button opens a registration dialog.



Register camera position for 488 nm

Place camera to the same focal plane for this filter, then choose how to register.

Current counter: 0

Current (mm): 0.000000 mm

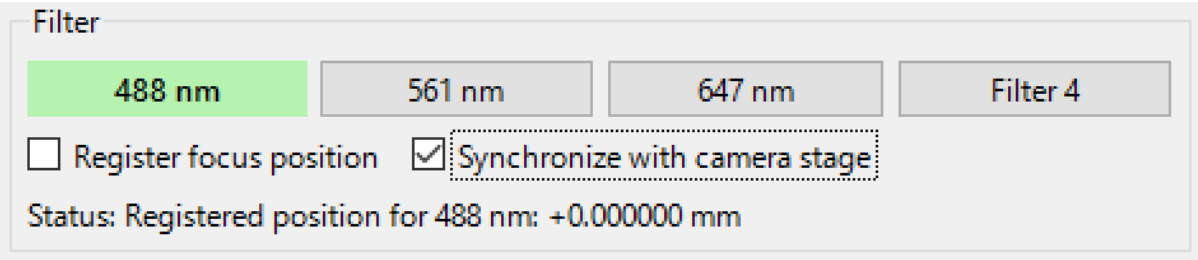
☒ Use current camera position
☐ Enter manual mm value (mm)

e.g. 0.012 (mm)

OK Cancel

You can set the position using the current stage position or by entering a value manually. Without moving the sample (i.e., keeping the same excitation plane), register the stage position for every filter. Based on the registered positions, the required camera-stage movement for focus correction during filter changes is calculated.

After registering all slots, be sure to uncheck **Register focus position**.



Filter

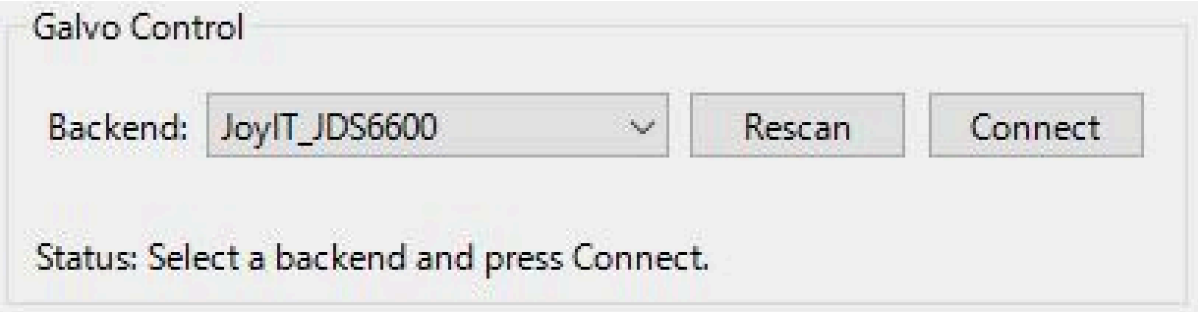
488 nm 561 nm 647 nm Filter 4

☐ Register focus position
 ☒ Synchronize with camera stage

Status: Registered position for 488 nm: +0.000000 mm

To enable focus correction, check **Synchronize camera stage**. When this is enabled, pressing a slot button moves the filter to the selected slot and simultaneously moves the camera stage for focus correction.

2-5 Galvano Control

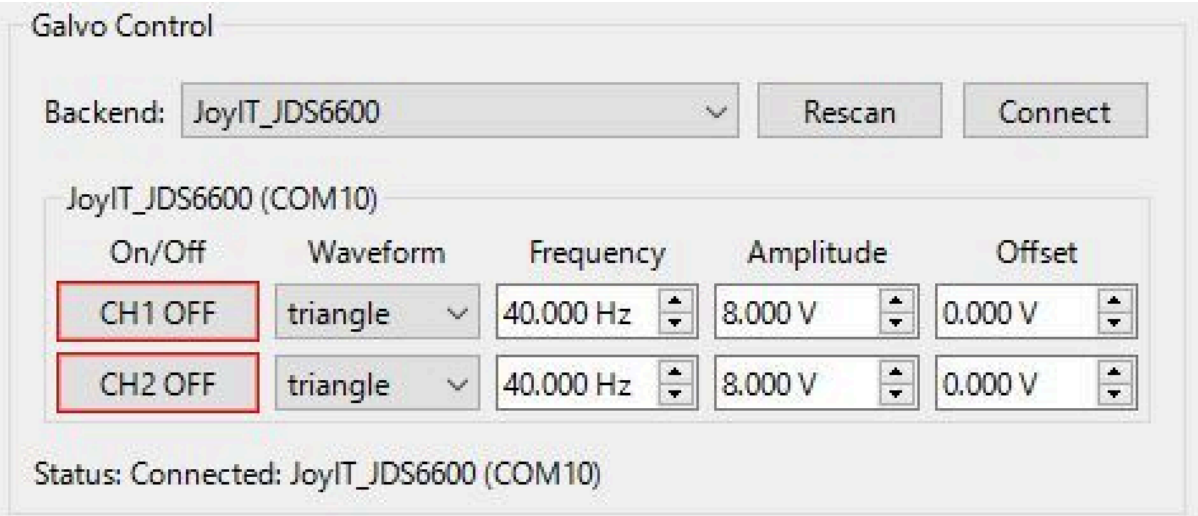


Galvo Control

Backend: JoyIT_JDS6600 ▼ Rescan Connect

Status: Select a backend and press Connect.

Select the backend for the connected pattern generator and press the **Connect** button. If no COM port is shown, check the connection to the PC and press the **Rescan** button. When the pattern generator is connected, the control panel will appear.



Galvo Control

Backend: JoyIT_JDS6600 ▼ Rescan Connect

JoyIT_JDS6600 (COM10)

On/Off	Waveform	Frequency	Amplitude	Offset
CH1 OFF	triangle ▼	40.000 Hz ▲▼	8.000 V ▲▼	0.000 V ▲▼
CH2 OFF	triangle ▼	40.000 Hz ▲▼	8.000 V ▲▼	0.000 V ▲▼

Status: Connected: JoyIT_JDS6600 (COM10)

The channel status button functions as an on/off toggle. To change the settings, enter the waveform, frequency, amplitude, and offset.

2-6 Data acquisition

Connect all devices. Set the camera exposure, laser power, movement parameters for the two stages, filter positions, and galvo settings. Place the sample and turn on the laser and galvo.

[For single-color imaging or the first track of multi-color imaging]

Focus on the sample. Use **Start (test)** or **Start (opposite)** in the stage control panel to move to the desired starting position, then press **Register Start Position** to save the start position.

In the camera panel, select the **Mode**, and if you want to use automatic return to the start position, check **auto-return**. Press the **Start Recording** button. When the stage reaches the end of the desired imaging range, press **Start Recording (Stop)** again to stop acquisition. A save dialog will appear; save the data to any location.

[For the second and subsequent tracks of multi-color imaging]

Confirm that the stages have returned to the start position and have stopped. If they have not returned, press **Go to Start Position**. Then replace the filter. Change the laser wavelength in the laser panel.

Check **Synchronize camera stage**, and when you switch the filter slot, focus adjustment is performed automatically.

Check the number of frames acquired in the first track, enable **Auto-stop at N frames**, and enter the same number.

Press the **Start Recording** button in the camera panel. Acquisition will stop automatically, and a save dialog will appear; save the data to any location.

If you want to add more tracks, repeat this procedure.

3. Troubleshooting

After connecting the laser, if you turn the physical key, the laser will not emit unless you press the ON/OFF button twice the first time.

The directions of Forward and Reverse may differ depending on the device, so always confirm the direction of movement.

If the stage behaves unexpectedly, you can stop it by pressing the button on the KDC101.

Be aware that when moving the stage in the negative direction, it first moves back significantly and then moves forward to the set position to reduce the effect of backlash.